Four gates before any science

Each one builds a world where the right answer is known, and asks what our operators return

1 Cartesian scheme · the time step

Setup a front in pure strain, flat grid
Truth G grows on a clock: exp(2αt)
Catches bad departure point or b interpolation

2 Native-grid metrics · the gradients

Setup an analytic field of lon, lat on the tile
Truth its gradient, in closed form
Catches wrong dxC, dyC, CS, SN or rotation

3 Discrete null · the identity itself

Setup advect a tracer, no mixing: DG/Dt = 2F
Truth slope 1 (got 0.95 / 0.76 → 1.00 / 0.98)
Catches the chain rule failing on the grid

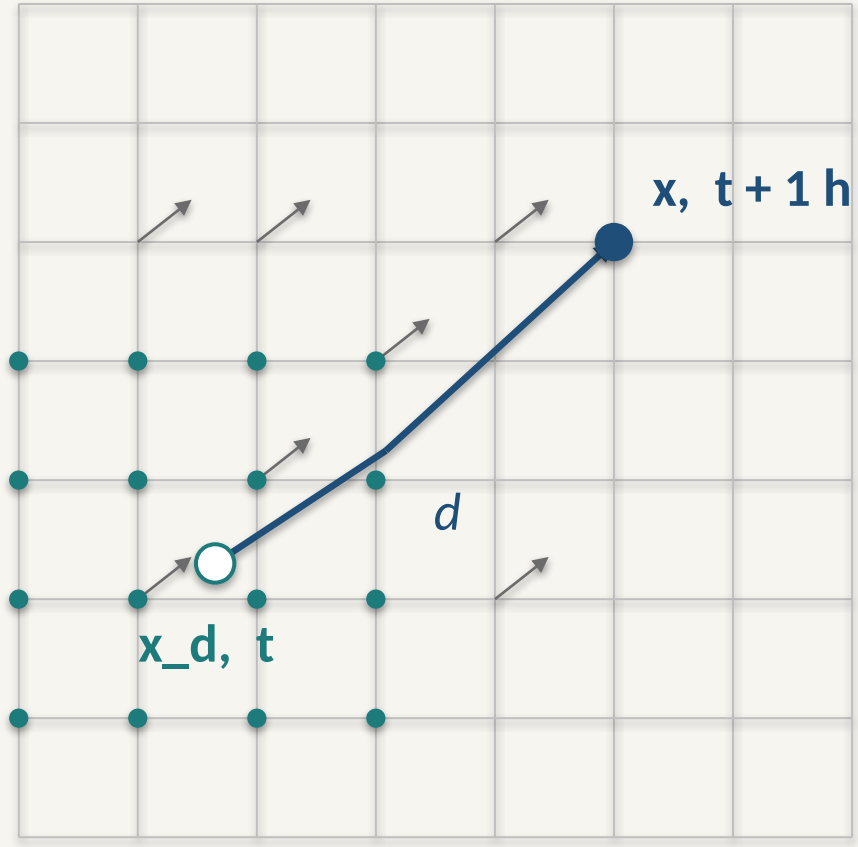
4 Interpolation bias · the zero

Setup carry G in a flow that only translates
Truth DG/Dt = 0 everywhere
Catches off-grid interpolation bias → error bar

All four run in M1. Gate 3 sets the baseline (0.981) every slope is quoted against; gate 4 the error bar.

The measured side: one difference along a parcel's path

DG/Dt = [G(x, t+1h) − G(x_d, t)] / 1h — two places, two times, one piece of water



Arrival a grid cell at the later hour. The parcel there is the one we follow.

Departure where that parcel was an hour earlier: x_d = x − d, with d = Δt · u at the path's midpoint, iterated three times.

Off-grid x_d almost never lands on a node, so b (and u, for d) are read from the 4 × 4 surrounding nodes by cubic interpolation.

Difference G at the arrival minus G at the departure, over one hour. That is DG/Dt following the water — the left side of the budget.

Gate 1 checks the path and the interpolation; gate 4 checks that pure translation gives zero.